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Dr. Romain A. Meyer

RESEARCH INTERESTS

Using a wide range of observational probes and facilities from the optical and infrared to the millimeter domain, I aim to characterise the properties of galaxies in the first billion years of the Universe, their role in cosmic hydrogen reionisation, and their co-evolution with their central supermassive black holes.

Keywords: Cosmic Reionisation, First Galaxies & AGN/SMBH, Multi-wavelength

EMPLOYMENT

Maître-Assistant (Research & Teaching Fellow) April 2023 - Present
Geneva Observatory, University of Geneva, Switzerland
Group of Prof. Pascal Oesch

Postdoctoral Researcher October 2020 - March 2023
Max Planck Institute for Astronomy, Heidelberg, Germany
Group of Dr. Fabian Walter

EDUCATION

PhD in Astrophysics 2017 - 2020
University College London, United Kingdom
Thesis: *The Role of Galaxies and Quasars in Reionising the High Redshift Intergalactic Medium*
Supervisor: Prof. Richard S. Ellis
Thesis official electronic version: <https://discovery.ucl.ac.uk/id/eprint/10112095/>

MSc in Physics 2015 - 2017
Ecole Polytechnique Fédérale de Lausanne, Switzerland
Thesis: *PSF Interpolation via Artificial Neural Networks*
Supervisors: Prof. Frédéric Courbin, Dr. Thibault Kuntzer

BSc in Physics 2012 - 2015
Ecole Polytechnique Fédérale de Lausanne, Switzerland / Erasmus at Imperial College, London
Thesis: *Cross-calibration of the Herschel SPIRE instruments*
Supervisor: Prof. Dave Clements, Dr. Rosalind Hopwood

PUBLICATIONS

77 (13 first-author) refereed papers published in major astronomical journals (A&A, MNRAS, ApJ, Nature, PASA, OJA) since 2018, totaling 3992 citations for an h-index of 34. A further 20 preprints under review, 2 proceedings publications and open source software releases. The full list of my publications can also be found on [NASA ADS](#) or on my [ORCID](#) profile.

GRANTS AND THIRD-PARTY FUNDING

NASA STScI Funding for JWST-GO Program #7208 (170'000 \$) December 2025
Administered by US Admin Co-PI Dr. Christina Williams
SNSF Scientific Exchange (19'200 CHF) October 2024
Funding for the organisation of the Saas-Fee Advanced Course 2025

AWARDED TELESCOPE TIME

PI-led Programmes:

Career total of > 600 hours on major observational facilities (e.g. JWST, VLT, ALMA, Gemini, NOEMA) with $> 5 - 10$ oversubscription factors.

JWST NIRSpec MSA - 108h	2026
"Mapping Reionisation in 3D in the COSMOS field (Cycle 5, #11371)"	
VLT/FORS2 - 84h	2026
"Mapping Reionisation in 3D in the COSMOS field (117.29KC)"	
ALMA Band 1 - 34.3h	2025
"Weighing the molecular gas reservoirs of UV-faint $z>6$ quasars" (2025.1.01221.S)	
JWST NIRSpec IFU SURVEY - 236.2h	2025
"THRIFTY: The High-Redshift Frontier survey" (Cycle 4, #7208)	
IRAM/NOEMA Band 3 - 30h	2024
"Characterising the ISM of a $z=6.5$ post-quasar galaxy" (W24EN)	
ALMA Band 1 - 32h	2024
"A survey of cold molecular gas in $z \sim 6.5$ quasars and their companion galaxies" (2024.1.00106.S)	
VLT/XShooter - 2.9h	2023
"Characterising a $z \sim 6$ gravitationally-lensed BAL quasar candidate" (113.26CY)	
JWST NIRSpec IFU - 10.5h	2022
"Characterizing the Source of Ionising Photons in the Epoch of Reionization" (#3767)	
Gemini (DDT) Flamingos-2 - 4.2h	2022
"Measuring the mass of the SMBHs powering two lensed $z \sim 6$ quasars" (DT-2022B-029)	
NOEMA Band 3 - 12h	2022
"Disentangling the nature of JWST ultra-high-redshift candidates with NOEMA" (W22EG)	
VLT/MUSE - 30h	2022
"A conclusive detection of galaxy overdensities around the first luminous quasars" (110.23UT)	
ALMA Band 6 - 14.6h	2021
"The Large-Scale Environment of the First Quasars" (2021.1.01557.S)	
ALMA Band 8/9 - 16.3h	2021
"Extreme Super-Eddington Star Formation in a Quasar Host at $z \sim 7$?" (2021.1.01350.S)	
NTT/EFOSC - 4 nights	2019
"A search for the missing gravitationally-lensed $z \sim 6$ quasars" (0104.A-0662(A))	

Selected Co-I Programmes:

ALMA LP - 113h	2025
"PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn" (PI: S. Shouws)	
JWST LP #5893 - 266h	2024
"COSMOS-3D: A Legacy Spectroscopic/Imaging Survey of the Early Universe" (PI: K. Kakiichi)	
JWST #5645 - 389h	2024
"A 3D view of the first QSOs: A JWST/NIRSpec survey program" (PI: E. Farina)	
VLT/MUSE LP - 147h	2023
"The cosmic Ecosystem of the first QSOs and Galaxies" (PI: E. Farina)	
ALMA - 49h	2023
"Delivering the Needed Large Samples of Extremely High SFR Sources at $z > 6$ to Characterize Early Stellar+Black-Hole Growth" (PI: R. Bouwens)	

“A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE)” (PI: F. Wang)

Additional Co-I Programmes: total of 409h with JWST, 290.1h with ALMA, 50.7h on the VLT (MUSE/ XShooter/FORS2), 22h on MEGARA/GTC, 15.6h on Subaru/PFS, 2×0.5 nights on Subaru/IRCS+LGS-AO, 6 nights on NTT, 92h on NOEMA, 38h on GEMINI GMOS, 19 orbits on HST, 333 ks on CHANDRA ACIS-S

STUDENTS SUPERVISION

Casey Hartmann, UNIGE MSc Semester Project	Spring 2026
Alexandre Pozzi, UNIGE MSc Thesis	February 2025 - January 2026
Lavinia Arpaia, UNIGE MSc Semester Project	Spring 2024
Alexandre Pozzi, UNIGE MSc Semester Project	Fall 2023
Xander Byrne, MPIA Summer Internship	Summer 2022
1 publication accepted in MNRAS (link here). Xander is now a PhD student at the University of Cambridge.	

THESIS JURIES AND EXAMINATIONS

External Examiner, MSc Thesis (A. Sawarkar), University of Geneva	June 2026
External Examiner, Geneva Highschool (CECG De Staël) Final Exams	June 2026

TEACHING

Guest Lecturer - Galaxies & Cosmology I & II (UNIGE)	2024 - present
MSc level Lectures on the Epoch of Reionisation / AGN / Observational Cosmology	
Teaching Assistant (UCL)	2019
Classical Mechanics	
Teaching Assistant (EPFL)	2016 - 2017
2nd year Physics Labs	
Teaching Assistant (EPFL)	2013 - 2016
General Physics for Engineers	

PROFESSIONAL SERVICE

Geneva Observatory - Postdoc Representative	2026 - present
Referee for Science	2026 - present
COSMOS-3D Collaboration - IGM & Reionisation WG Lead	2025 - present
Expert Panel Member - ESO Observing Programmes Committee	2025
Saas-Fee Advanced Course 2025 Organiser (website)	2025
EAS 2024 Special Session 1 Co-Organiser and Chair	2024
EPFL-UNIGE Journal Club Organiser	2023 - 2025
Galaxy Coffee Seminar Organiser, MPIA, Heidelberg	2021 - 2022
Walter Group Meetings Organiser, MPIA, Heidelberg	2021 - 2022
Referee for ApJ/AJ	2021 - present
Referee for MNRAS	2020 - present
Astronomy PhD Student Representative, UCL	2017- 2020

OBSERVING AND TECHNICAL EXPERIENCE

- **Observing experience:** 3 nights on Keck/MOSFIRE, 3 nights on Keck/DEIMOS, 4 nights on NTT/ EFOSC2, 3 nights (remote) on MEGARA/GTC
- **Data reduction & analysis:** optical/NIR multi/long-slit spectroscopy (DEIMOS/Keck, MOSFIRE/Keck, EFOSC2/NTT, GMOS/Gemini, Flamings2/Gemini, VLT/XShooter), optical IFU (MUSE/VLT), mm interferometry (ALMA/NOEMA), JWST NIRISS/NIRCam WFSS and imaging data, JWST NIRSpec IFU
- **Current main developer** and maintainer of [interferopy](#); an open-source, public python library for radio/mm interferometry data analysis.
- **Computing:** Python, C++, Fortran, IDL
- **Astronomy software:** CASA, GILDAS, ESORex, ESOReflex, PyeIt, DS9, Topcat, SourceExtractor, VPFit, GalSim, GALFIT, Grizli, MIRAGE, JWST pipeline
- **Summer schools:** European Radio Interferometry School 2019, Chalmers, Gothenburg / Astrostat 2021, Penn State (Virtual) / ISM Galaxies 2021, Marseille (Virtual)

INVITED AND CONTRIBUTED TALKS

Invited Talks and Colloquia:

Miracles of the Early Universe - Geneva, Switzerland	June 2025
<i>Connecting the Sources of Reionisation to the IGM topology</i>	
Hamburg Observatory Colloquium - Hamburg	January 2025
<i>The ALMA View of Quasars in the First Billion Years</i>	
Galaxy Evolution Circle - LAM, Marseille	December 2024
<i>Can we solve Reionisation on the scales of individual ionised bubbles?</i>	
Cake Talk - DAWN, Copenhagen	May 2024
<i>Probing early galaxy formation with an unbiased sample of $6.8 < z < 9$ [OIII] emitters</i>	
Galaxies Discussion Group - KICC, Cambridge	October 2022
<i>A resolved view of $z > 6$ quasar host galaxies</i>	
Extragalactic Seminar - UCL, London	October 2022
<i>A 100-300 pc view of $z > 6.5$ quasar host galaxies</i>	
Young Astronomers on AGN 2022 - DIPC, San Sebastian	October 2022
<i>Quasar Host Galaxies at Cosmic Dawn</i>	
Astronomy Seminar - University of Southampton	March 2022
<i>Quasar Host Galaxies at Cosmic Dawn</i>	
Königstuhl Colloquium - MPIA, Heidelberg	April 2021
<i>Finding the sources of reionisation</i>	
Galaxies Discussion Group - KICC, Cambridge	April 2020
<i>IGM-galaxy cross-correlations during Reionization</i>	
MPIA Galaxy Coffee - MPIA, Heidelberg	November 2019
<i>Probing reionisation with cross-correlations of galaxies and the Lyman-α forest at $z \sim 6$</i>	
London Cosmology Discussion Meeting - RAS, London	November 2018
<i>Probing the epoch of reionisation with cross-correlations of high-redshift galaxies and the IGM transmission</i>	

Contributed Talks:

MIAPbP Workshop: Continuing the JWST revolution - Munich, DE	June 2026
<i>Towards a 3D view of Reionisation</i>	
Massive Black Holes across Cosmic Time - Cambridge, UK	September 2025
<i>Discovery of two $z \sim 6.5$ post-quasar galaxies in large ionised bubbles</i>	
EREBUS Collaboration Meeting - Bologna, Italy	June 2025

<i>A MUSE Archival View of $z \sim 6$ Quasar Environments</i>	
Miracles of the Early Universe - Geneva, Switzerland	June 2025
<i>Towards a 3D view of Reionisation: Connecting the Sources to the IGM topology</i>	
Galaxy Origins in the JWST era - Toledo, Spain	May 2025
<i>Discovery of two $z \sim 6.5$ post-quasar galaxies in large ionised bubbles</i>	
Views on the multi-phase ISM in galaxies - Bologna, Italy	September 2024
<i>Pushing ALMA to the limit: 140-pc resolution [CII] and continuum observations of a $z=6.6$ quasar-galaxy merger</i>	
Observing and Simulating Galaxy Evolution - Ascona, Switzerland	August 2024
<i>Constraining early galaxy formation with an unbiased sample of $6.8 < z < 9$ [OIII] emitters</i>	
EAS 2023 - Krakow, Poland	July 2023
<i>Contributed talks in Symposium 12, Special Session 22 & Poster Presentation in Symposium 8</i>	
First Light Conference, MIT, Boston	June 2023
<i>A complete census of [OIII]+Hβ emitters at $6.7 < z < 9$ with JWST</i>	
Exploring the evolving Universe (RSE@70+) - IoP, London	June 2023
<i>A complete census of [OIII]+Hβ emitters at $6.7 < z < 9$ with JWST</i>	
From Stars to Galaxies II - Gothenburg, Sweden	June 2022
<i>Hyper-resolution observations of $z \sim 6.8$ quasar hosts</i>	
Learning the high-z Universe - Sazerac Online Conferece	February 2022
<i>Learning from Quasars with VAEs</i>	
Quasars and Galaxies Through Cosmic Time - Chile / Online	February 2022
<i>Constraining galaxy overdensities around three $z \sim 6.5$ quasars with ALMA and MUSE</i>	
RAS Specialist Day - RAS London / Online	January 2022
<i>Measuring the contribution of $z \sim 6$ galaxies to reionisation with galaxy-IGM cross-correlations</i>	
EAS 2021 - Session 2 - Leiden Observatory / Online	July 2021
<i>High-redshift quasar hosts viewed by ALMA and NOEMA</i>	
EAS 2021 - Session 7 - Leiden Observatory / Online	July 2021
<i>Measuring escape fractions at $z \sim 6$ with quasar spectra</i>	
SAZERAC 2020 - Online	July 2020
<i>Measuring the ionising photon escape fraction of $z \sim 6$ galaxies</i>	
EAS 2020 - Leiden Observatory / Online	June 2020
<i>Discovery of a double-peaked Lyman alpha emission in a galaxy at $z = 6.802$</i>	
KICC 10th Anniversary Symposium - KICC, Cambridge	September 2019
<i>Probing the epoch of reionisation with cross-correlations of high-redshift galaxies and the IGM transmission</i>	
EAS 2019 (Session 2) - Lyon, France	June 2019
<i>Evidence for quasar evolution: rest-frame UV broad lines shifts at $1.5 < z < 7.5$</i>	
EAS 2019 (Session 3) - Lyon, France	June 2019
<i>A new route to the contribution to reionisation of subluminous $z \sim 6$ galaxies</i>	
What matters between galaxies? - Abbazia di Spineto, Firenze, Italy	June 2019
<i>Metal-tracing the sources of reionisation</i>	
IGM2018: Revealing Cosmology and Reionization History with the Intergalactic Medium - Kavli IPMU, Kashiwa, Japan	September 2018
<i>Faint galaxies reionising the IGM at $z \sim 5$: metal-tracing the sources of reionisation</i>	

OUTREACH

Supervision of 3 high-school interns (1 week)	May 2025
Geneva Observatory	
Public Talk (French): “La Fin des Âges Sombres”	April 24, 2026
Société Astronomique du Valais Romand	
Public Talk (French): “La Fin des Âges Sombres”	February 7, 2025

Société d'Astronomie du Nord Vaudois

Geneva Observatory Visits

2024 - present

University of Geneva - Visits for general public (adults and children)

Astronomy On Tap London

2020

Co-founder and online host (during Covid-19)

UCL Diploma Club Seminar Organiser

2019 - 2020

Seminar series for graduates of the UCL evening course open to the general public

ORBYTS: Researcher in schools program

2017 - 2019

Outreach programs for highschool pupils of 10 weeks of lectures and research on real data

Mid-Kent Astronomical Society

January 31 2020

Outreach talk: Galaxies in the First billion years

Bounce Back RAS200 Project (HMP Brixton)

November 4 2019

Outreach to inmates of a London prison - with the RAS and the Bounce Back foundation

UCL Future Frontiers Event Careers Networking Event

July 2019

Day of networking/career advice to London pupils from underprivileged background

International Day of Light - UCL

May 2018

Day of outreach shows and demonstrations for secondary school pupils

PRESS COVERAGE

- November 2022: “Dans les coulisses des données du télescope spatial James Webb”: radio interview and web article with the Swiss public broadcasting organisation RTS on the analysis of early JWST data ([article and radio interview in French](#))
- July 2021: “Cosmic Dawn occurred 250 to 350 million years after Big Bang” press release linked to preprint arxiv.org/abs/2104.08168. Covered by the [BBC press](#)/TV/radio, [the Guardian](#), the [Daily Mail](#), the [Evening Standard](#) and other specialised media. Live interview with BBC World News TV (24/06/21).
- July 2020: EAS press release “Discovery of a luminous galaxy reionizing the intergalactic medium 13 billion years ago”, covered by [specialised internet media](#).